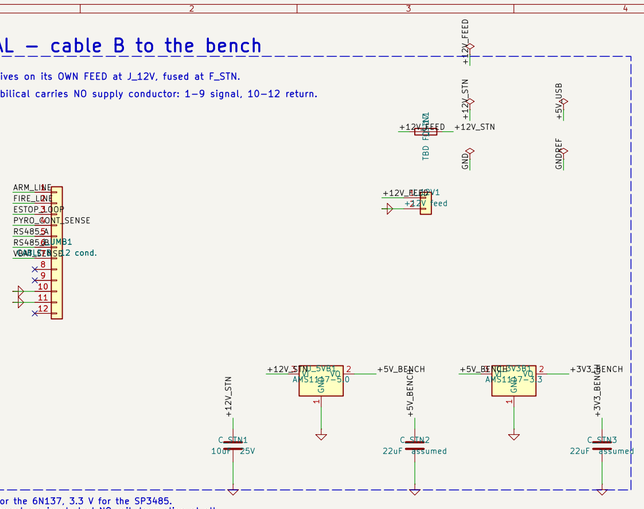
1. UMBILICAL — cable B to the bench

The station supply arrives on its OWN FEED at J_12V, fused at F_STN.

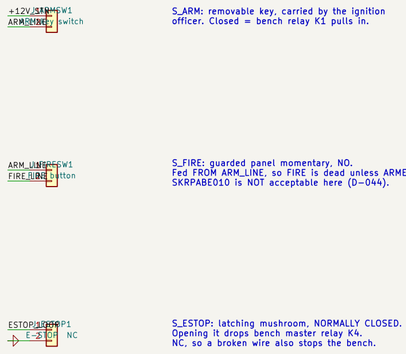
The 12-conductor umbilical carries NO supply conductor: 1–9 signal, 10–12 return.



Bench-side rails: 5 V for the 6N137, 3.3 V for the SP3485.
C_STN1–4 added: this board previously had NO rail decoupling at all.
22 uF values are ASSUMED from the AMS1117 stability note — confirm (O-20).

2. CONTROLS — mechanical, panel-mounted

All four are PANEL controls wired to these terminals, not PCB parts.



S_ARM: removable key, carried by the ignition officer. Closed = bench relay K1 pulls in.

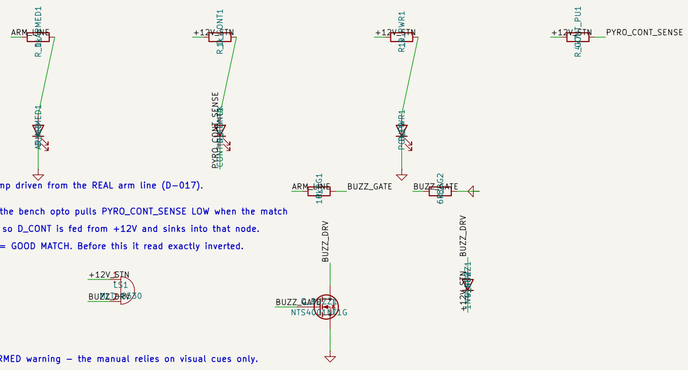
S_FIRE: guarded panel momentary, NO.
Fed FROM ARM_LINE, so FIRE is dead unless ARMED.
SKRPABE010 is NOT acceptable here (D-044).

S_ESTOP: latching mushroom, NORMALLY CLOSED.
Opening it drops bench master relay K4.
NC, so a broken wire also stops the bench.

H-2 FIX: the CONTINUITY test button is REMOVED.
As drawn it shorted PYRO_CONT_SENSE to ground, which is the lamp-good state, so the test reported a good match whether or not one was connected. A test that cannot fail is worse than no test.
Sensing is continuous and needs no button to initiate it.

OPEN DESIGN QUESTION: because sensing is continuous, current flows through the e-match whenever the bench is powered. Gating that needs a decision — see the audit, H-2.

3. INDICATORS & AUDIO



ARMED lamp driven from the REAL arm line (D-017).

H-1 FIX: the bench opto pulls PYRO_CONT_SENSE LOW when the match conducts, so D_CONT is fed from +12V and sinks into that node.
LAMP LIT = GOOD MATCH. Before this it read exactly inverted.

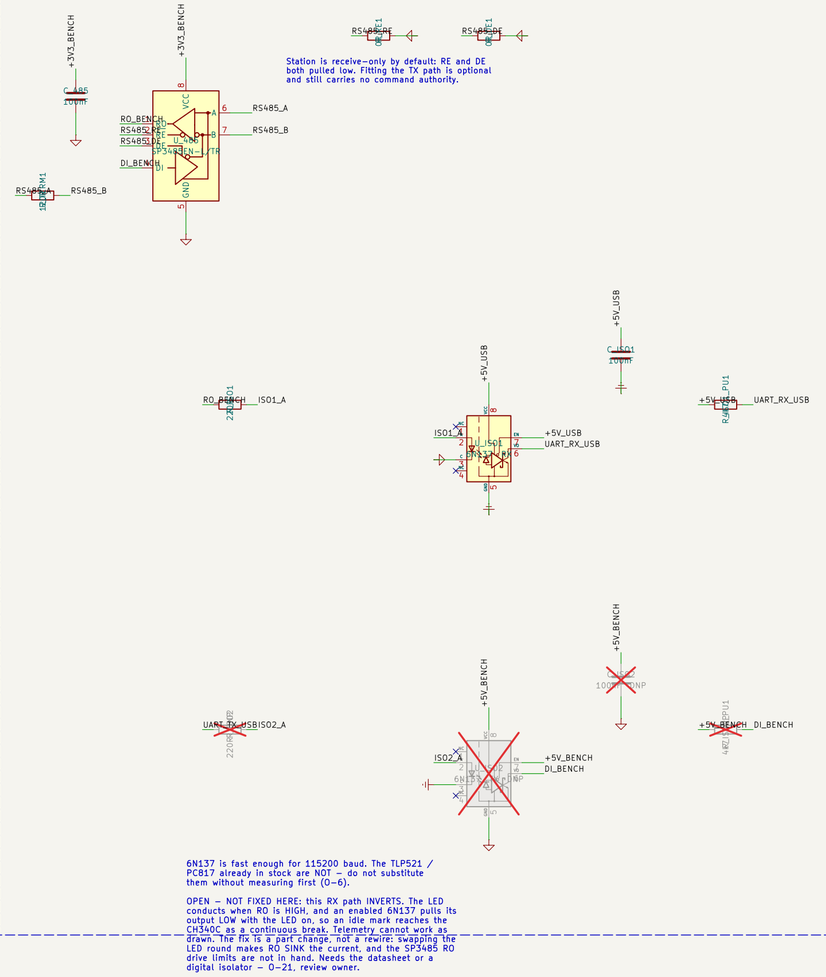
Audible ARMED warning — the manual relies on visual cues only.

4. TELEMETRY — RS-485 in, isolated, out to USB

ISOLATION BARRIER down the middle of this block.

LEFT = bench-referenced (GND). RIGHT = laptop side (GNDREF).

No isolated DC-DC: the laptop's USB supplies the isolated side (D-056).

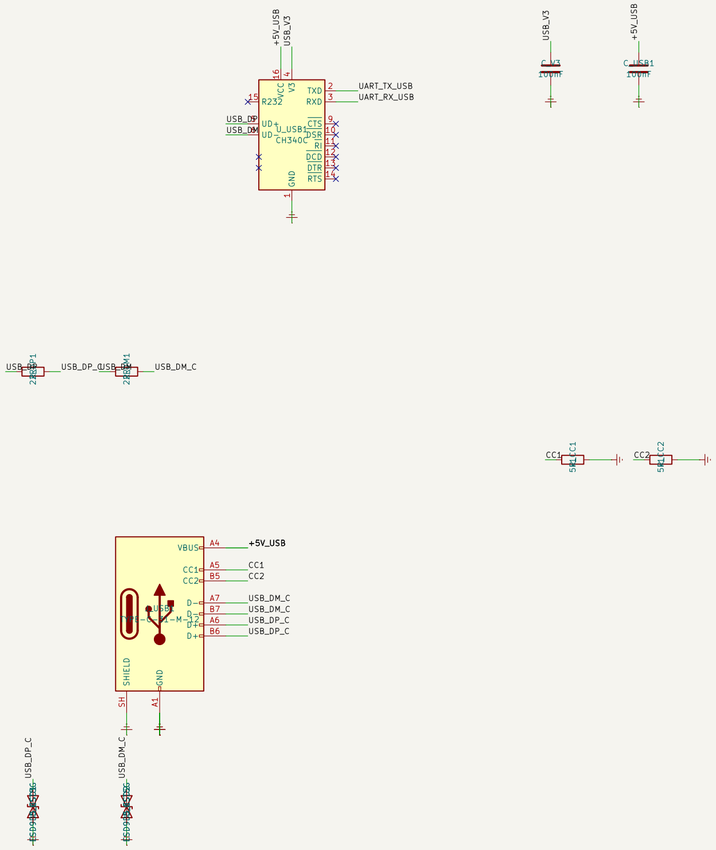


Station is receive-only by default: RE and DE both pulled low. Fitting the TX path is optional and still carries no command authority.

6N137 is fast enough for 115200 baud. The TLP521 / PC817 already in stock are NOT — do not substitute them without measuring first (O-6).

OPEN — NOT FIXED HERE: this RX path INVERTS. The LED conducts when RO is HIGH, and an enabled 6N137 pulls its output LOW with the LED on, so an idle mark reaches the CH340C as a continuous break. Telemetry cannot work as drawn. The fix is a part change, not a rewire: swapping the LED round makes RO SINK the current, and the SP3485 RO drive limits are not in hand. Needs the datasheet or a digital isolator — O-21, review owner.

5. USB-C — laptop side, GNDREF domain



CC pull-downs 5k1 mark this as a UFP sink.
Data series 22R, one ESD diode per line.

6. MEASUREMENT OUTPUT — works with no software at all

PATH 2 IS DEFERRED FOR REV A. The SD log is the measurement.



J_DAQ IS NOT FITTED IN REV A (O-19).
It was wired to umbilical pins 8/9, which reach NOTHING on the bench: no buffer, no source. A connector present at both ends of a dead pair reads as a working analog backup, so it is unpopulated and its pins are flagged rather than left inviting a DAQ.
Pin 3 also exported bench GND to the laptop bench, which bridges the isolation barrier whenever a mains-powered DAQ shares it.
Loss of path-2 redundancy for Rev A is recorded in REQUIREMENTS.md.



Battery voltmeter: the pre-test >= 12 V check (manual p. 12), read without walking to the bench.

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_station.py
TWO GROUND DOMAINS: GND = bench-referenced, GNDREF = laptop/USB. Never joined.
PRELIMINARY — NOT APPROVED. ARM, FIRE and E-STOP are mechanical, panel-mounted.

Ignitia Rocket Lab	
Sheet: /	
File: talavera-station.kicad_sch	
Title: TALAVERA v1 — Control Station Board	
~~Size: A2~~ ~~Date: 2026-11-06~~	**~~Rev: A~~**
KiCad E.D.A. 10.0.5	Id: 1/1

TBD values await O-2 (load cell) and O-6 (isolator part). ARM / FIRE / E-STOP carry no firmware.